

26 By Faraday's Law

$$\begin{aligned}\varepsilon_{\text{induced}} &= \frac{d}{dt} NBA = NB \frac{dA}{dt} \approx NB \left(\frac{\Delta A}{\Delta t} \right) \\ &= (1)(2.5 \times 10^{-2}) \frac{\pi \left(\frac{3.0 \times 10^{-2}}{2} \right)^2 - 0}{0.050} \\ &= 3.5 \times 10^{-5} \text{ V} \\ P &= \frac{\varepsilon^2}{R} = 2.5 \times 10^{-8} \text{ W}\end{aligned}$$

By Lenz's Law and right hand grip rule,
since the change is reduction of area of inward magnetic field,
direction of induced current is to produce inward magnetic field,
current flows out to a circuit "external" to XY through X
so X is higher potential.

27 Eliminate A and B: $P = I^2 R$ so should be sine-squared instead of $|\sin|$